

PH1101: Physics-I

Dr. Alpana Nayak

(Department of Physics)

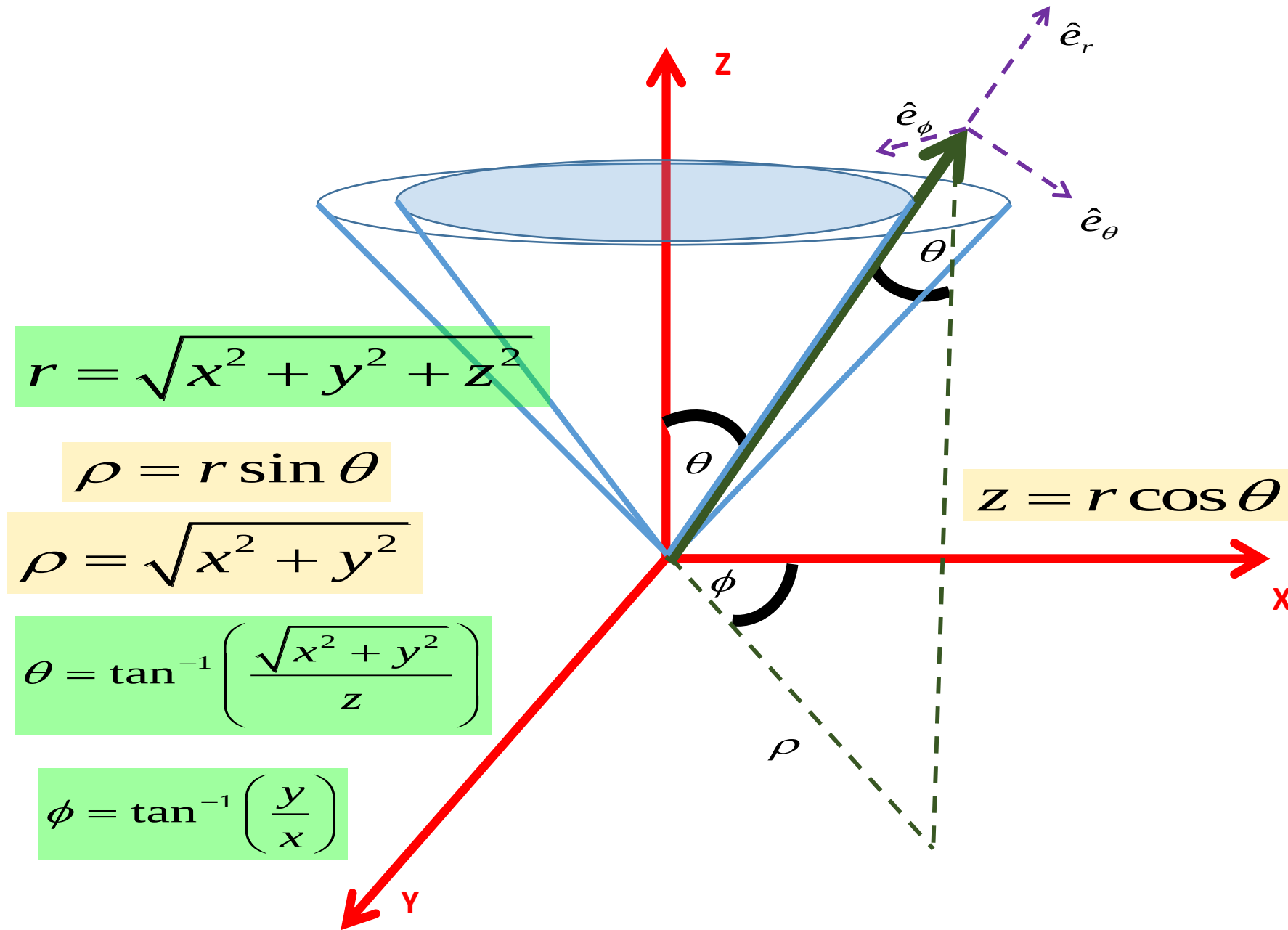
Office:
Block-IV (Room 123)

Email: anayak@iitp.ac.in

11/08/2025

Orthogonal Co-ordinate systems

Spherical Polar Coordinate System



Transformation of Coordinates

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right)$$

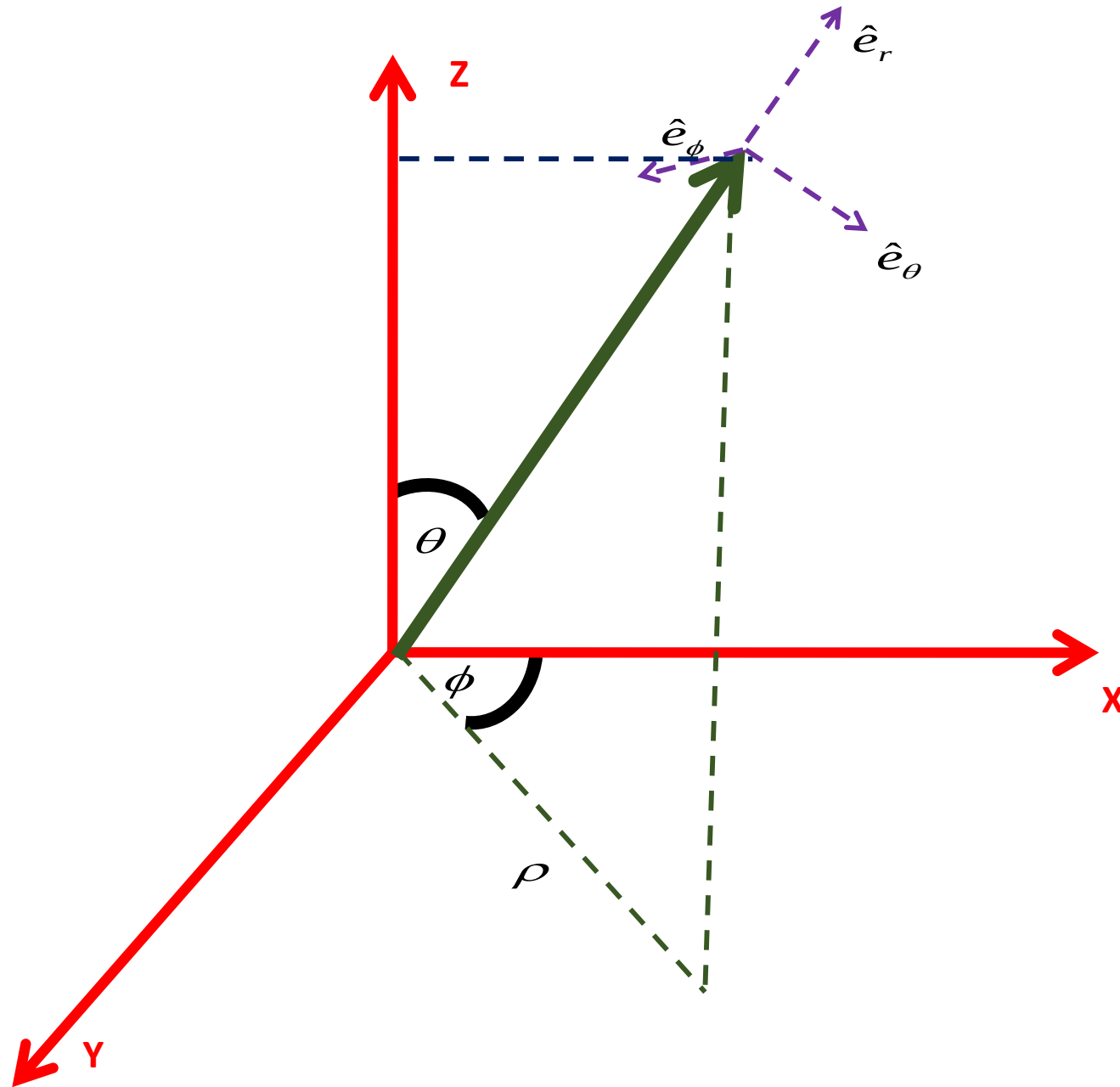
$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$x = r \sin \theta \cos \phi$$

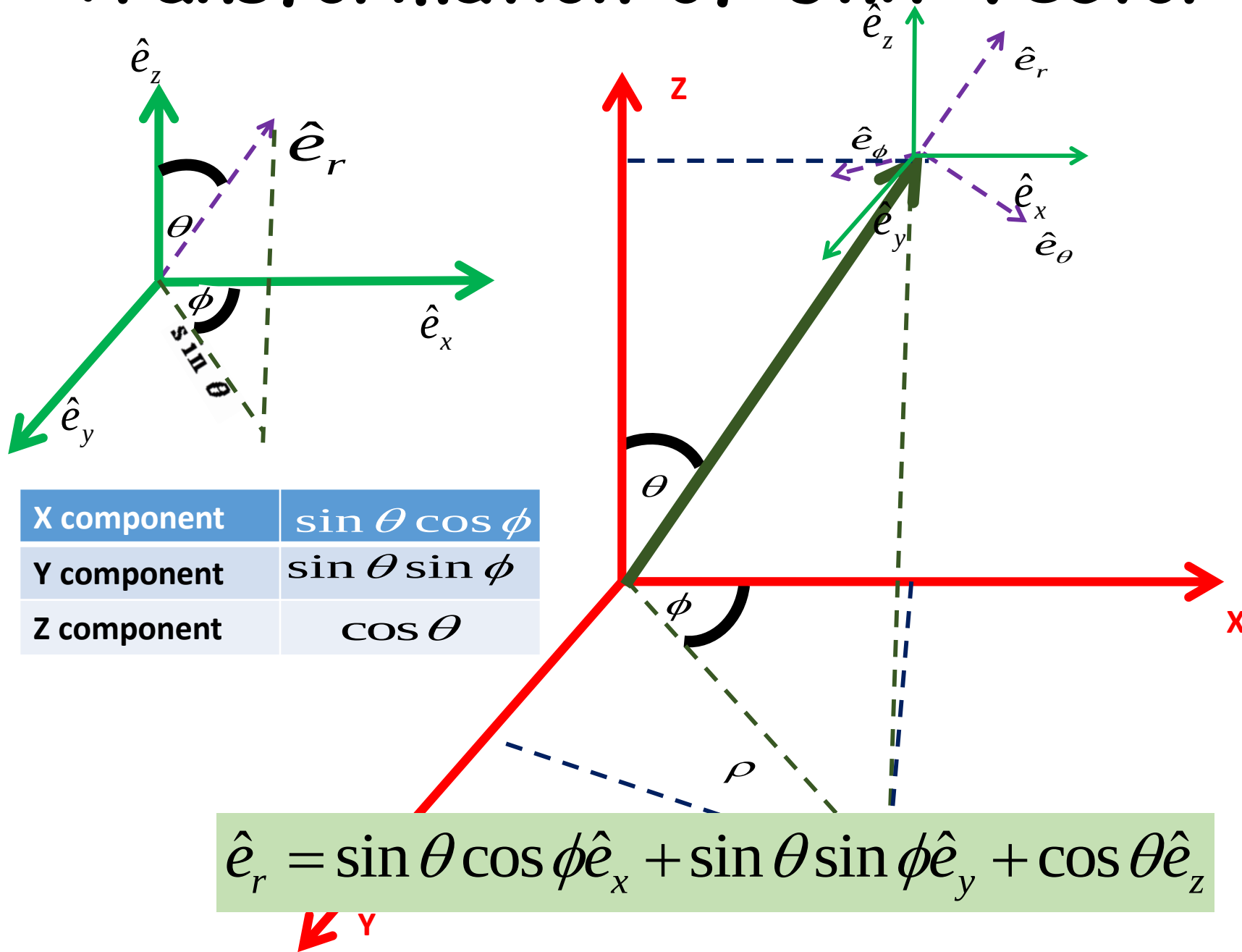
$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

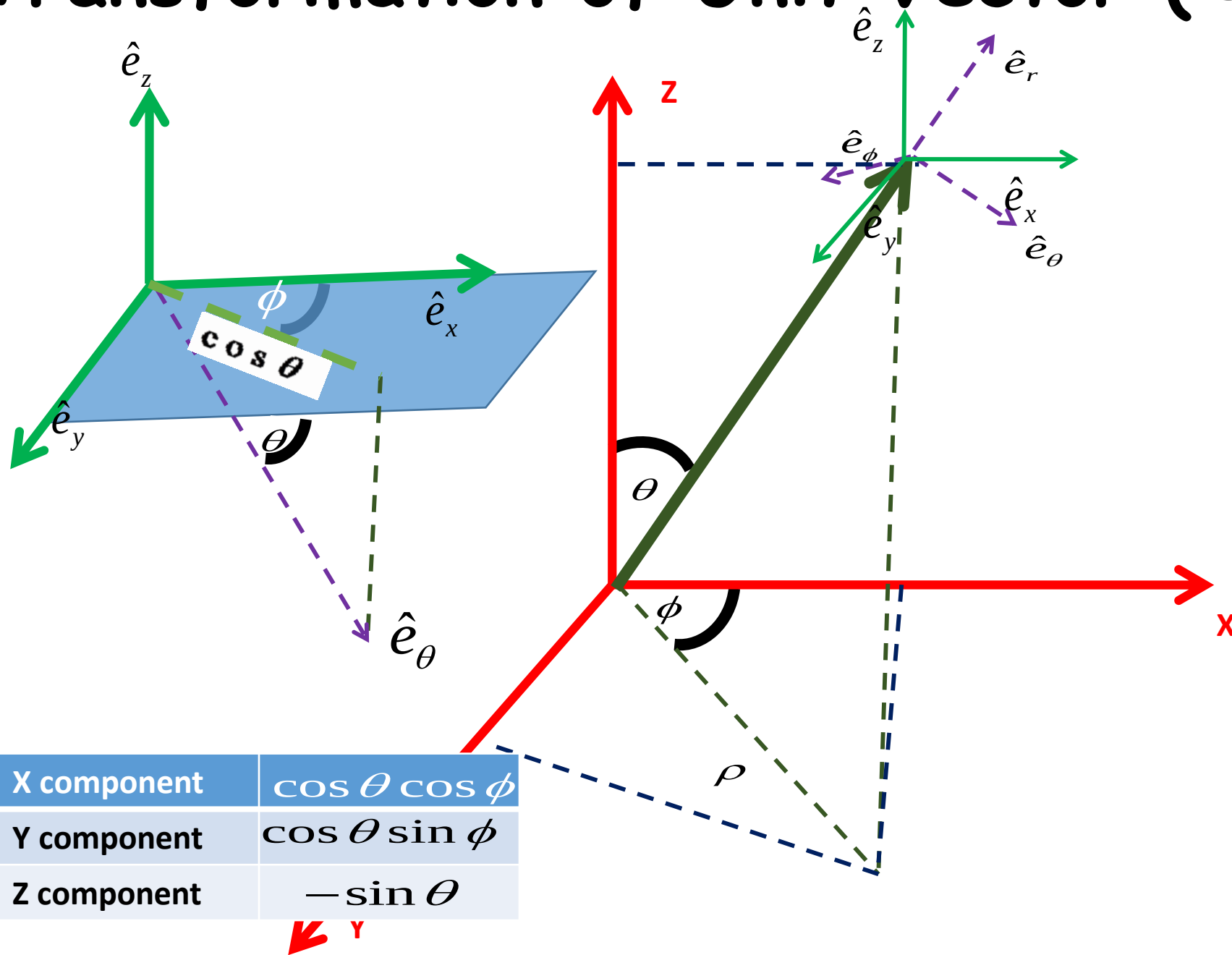
Transformation of Unit Vectors



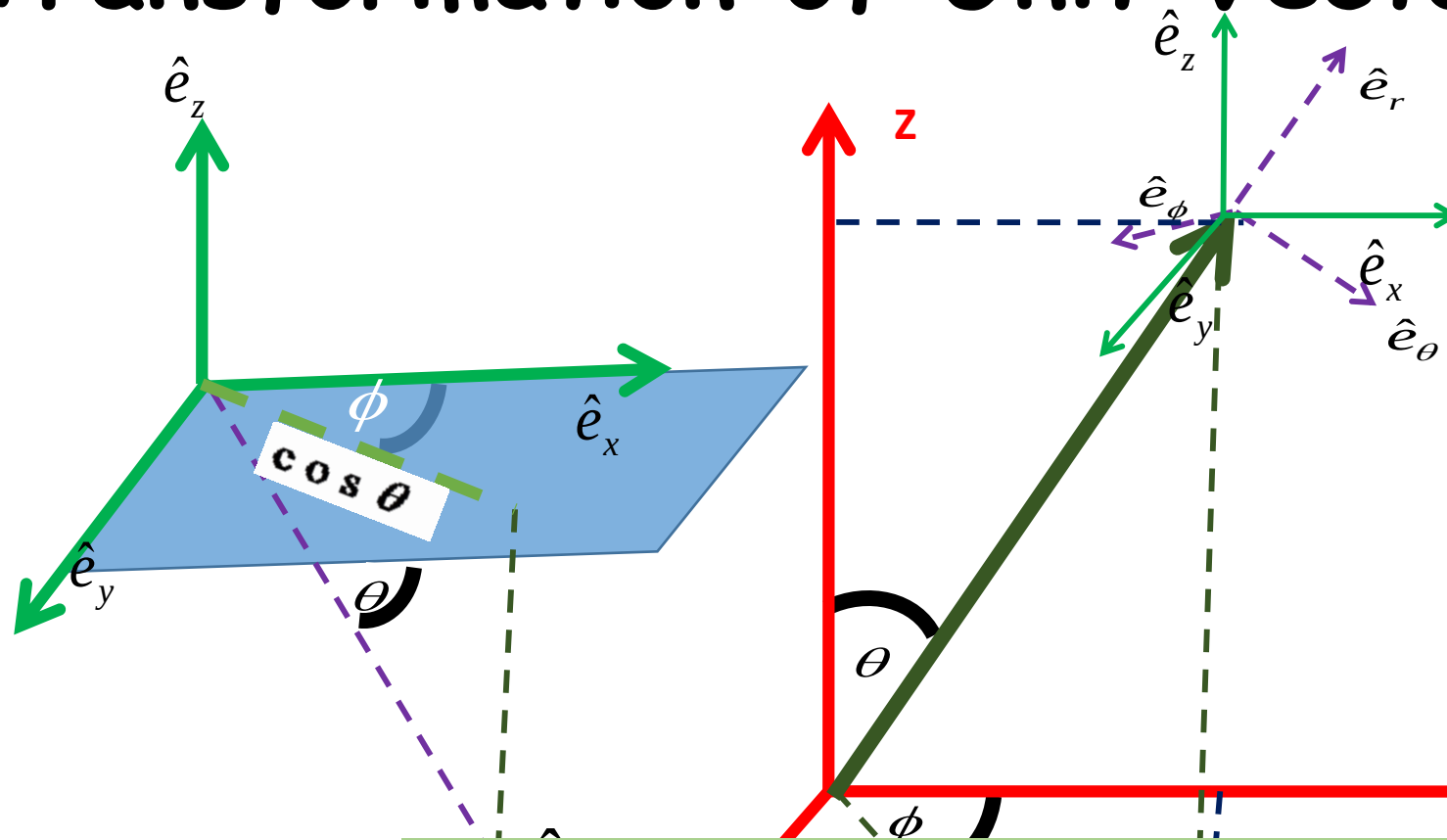
Transformation of Unit Vector ($\hat{e}_r$)



Transformation of Unit Vector ($\hat{e}_\theta$)



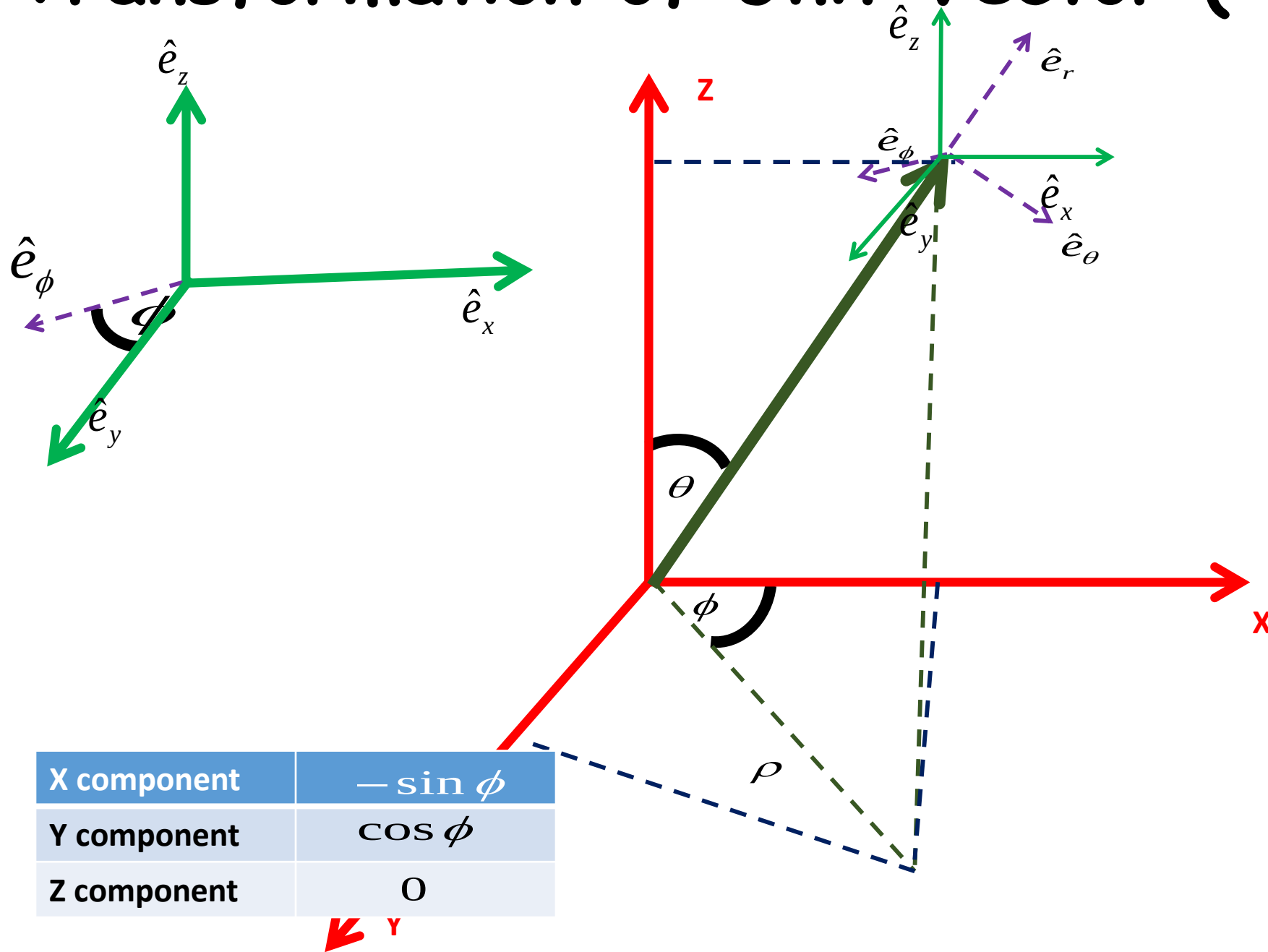
Transformation of Unit Vector ($\hat{e}_\theta$)



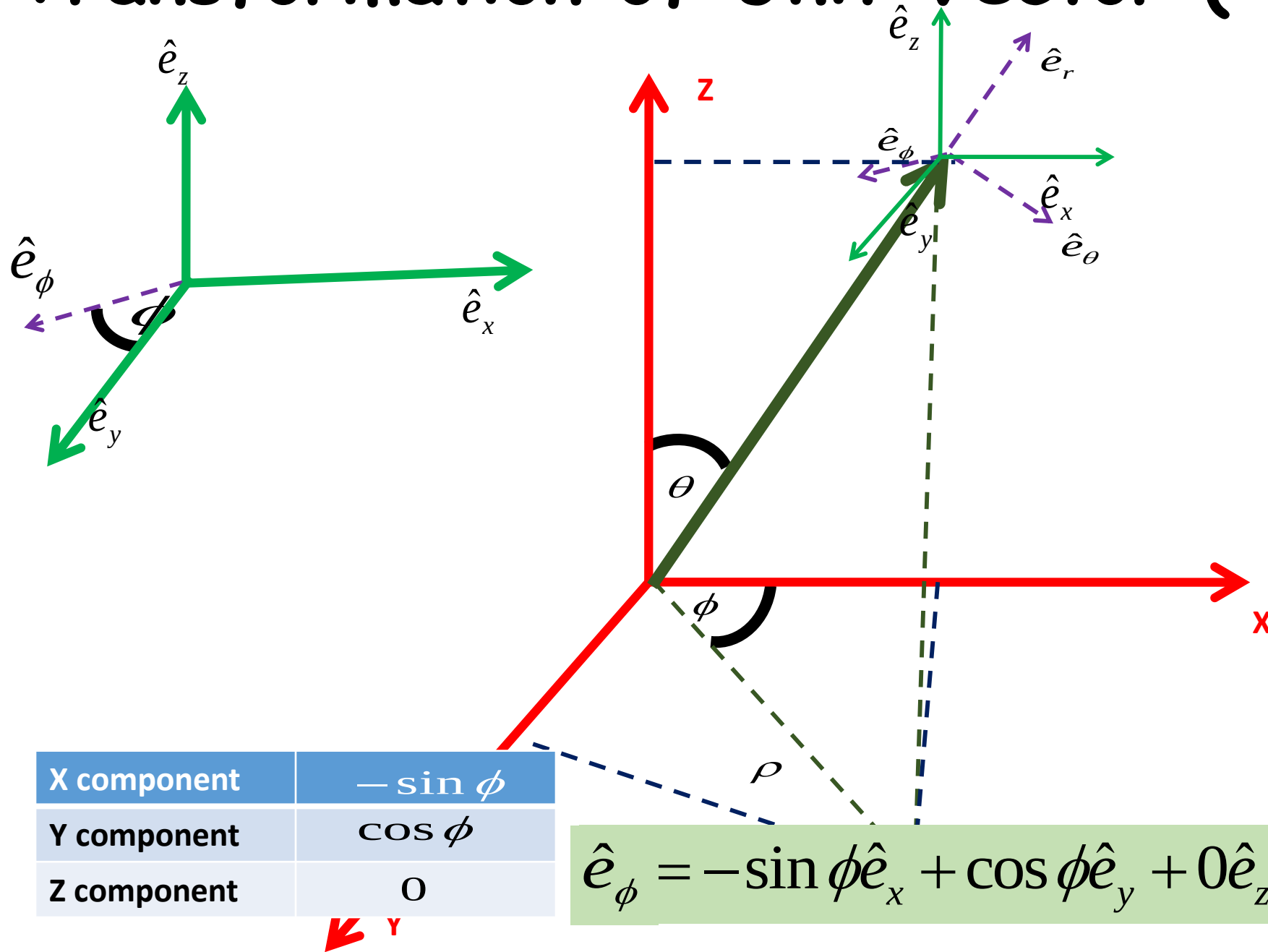
$$\hat{e}_\theta = \cos \theta \cos \phi \hat{e}_x + \cos \theta \sin \phi \hat{e}_y - \sin \theta \hat{e}_z$$

X component	$\cos \theta \cos \phi$
Y component	$\cos \theta \sin \phi$
Z component	$-\sin \theta$

Transformation of Unit Vector ($\hat{e}_\phi$)



Transformation of Unit Vector ($\hat{e}_\phi$)



Transformation of Unit Vectors (Matrix notation)

$$\hat{e}_r = \sin \theta \cos \phi \hat{e}_x + \sin \theta \sin \phi \hat{e}_y + \cos \theta \hat{e}_z$$

$$\hat{e}_\theta = \cos \theta \cos \phi \hat{e}_x + \cos \theta \sin \phi \hat{e}_y - \sin \theta \hat{e}_z$$

$$\hat{e}_\phi = -\sin \phi \hat{e}_x + \cos \phi \hat{e}_y + 0 \hat{e}_z$$

$$\begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \\ \hat{e}_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \\ \hat{e}_z \end{bmatrix}$$

Transformation of Unit Vectors (Matrix notation)

$$\begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \\ \hat{e}_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \\ \hat{e}_z \end{bmatrix}$$

$$\begin{bmatrix} \hat{e}_x \\ \hat{e}_y \\ \hat{e}_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \\ \hat{e}_\phi \end{bmatrix}$$

Derivatives of Unit Vectors

$$\begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \\ \hat{e}_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \\ \hat{e}_z \end{bmatrix}$$

$$\frac{d\hat{e}_r}{dr} = 0$$

$$\frac{d\hat{e}_r}{d\theta} = \hat{e}_\theta$$

$$\frac{d\hat{e}_r}{d\phi} = \sin \theta \hat{e}_\phi$$

$$\frac{d\hat{e}_\theta}{dr} = 0$$

$$\frac{d\hat{e}_\theta}{d\theta} = -\hat{e}_r$$

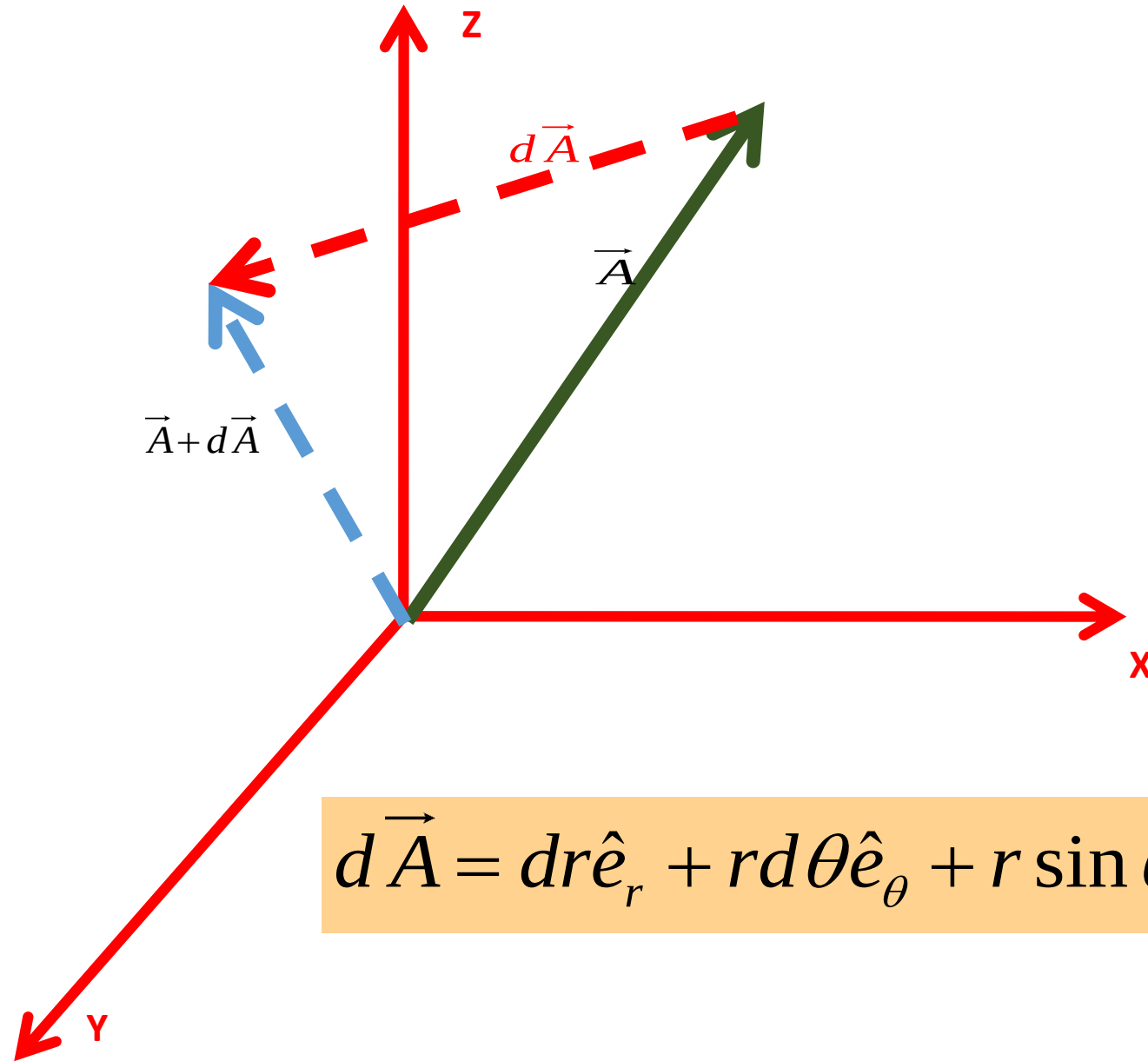
$$\frac{d\hat{e}_\theta}{d\phi} = \cos \theta \hat{e}_\phi$$

$$\frac{d\hat{e}_\phi}{dr} = 0$$

$$\frac{d\hat{e}_\phi}{d\theta} = 0$$

$$\frac{d\hat{e}_\phi}{d\phi} = -\cos \theta \hat{e}_\theta - \sin \theta \hat{e}_r$$

Infinitesimal line element

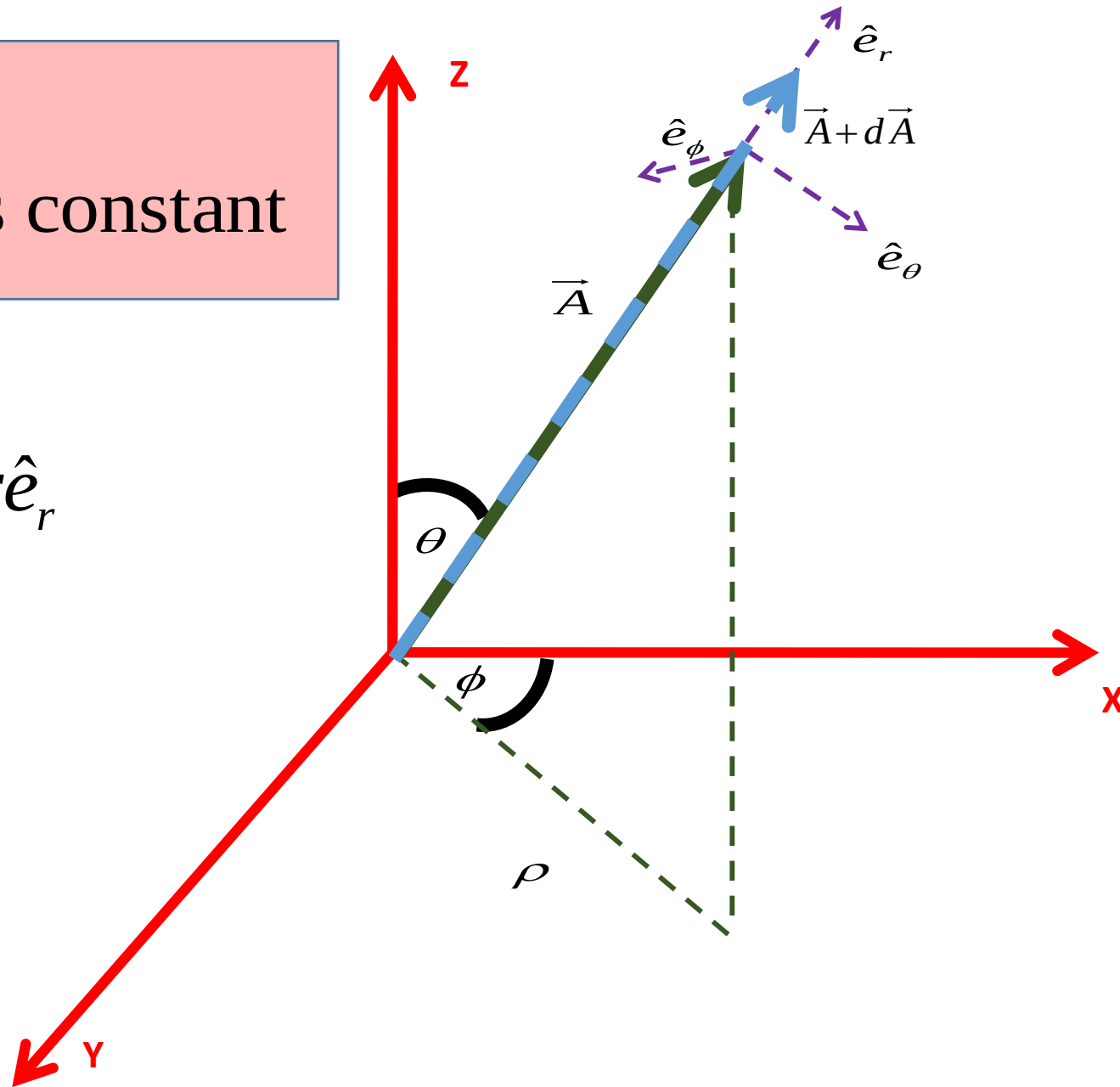


$$d\vec{A} = dr\hat{e}_r + r d\theta\hat{e}_\theta + r \sin\theta d\phi\hat{e}_\phi$$

Infinitesimal line element

Case 1:
 θ and ϕ is constant

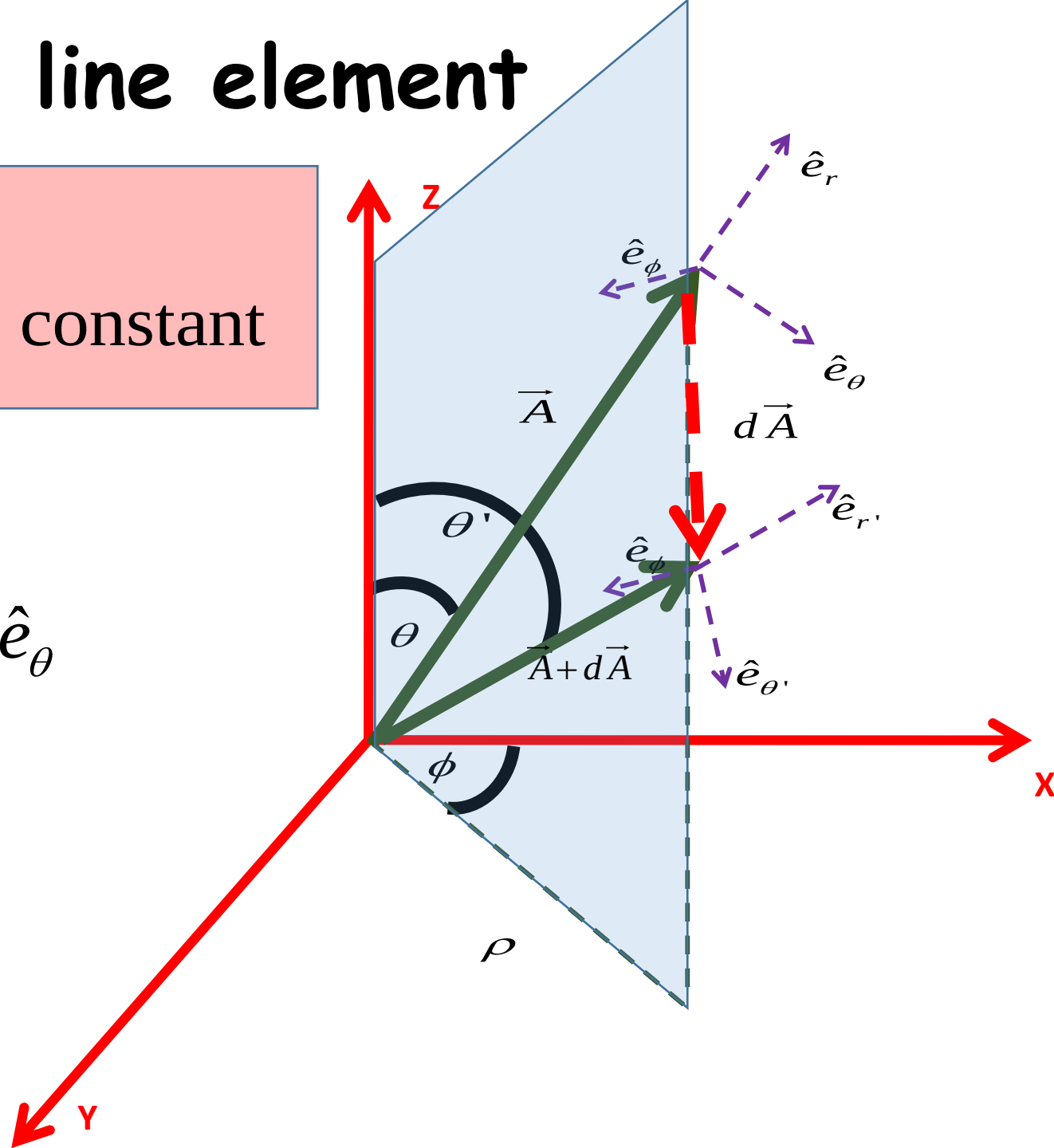
$$d\vec{A} = dr\hat{e}_r$$



Infinitesimal line element

Case 2:
 r and ϕ is constant

$$d\vec{A} = r d\theta \hat{e}_\theta$$

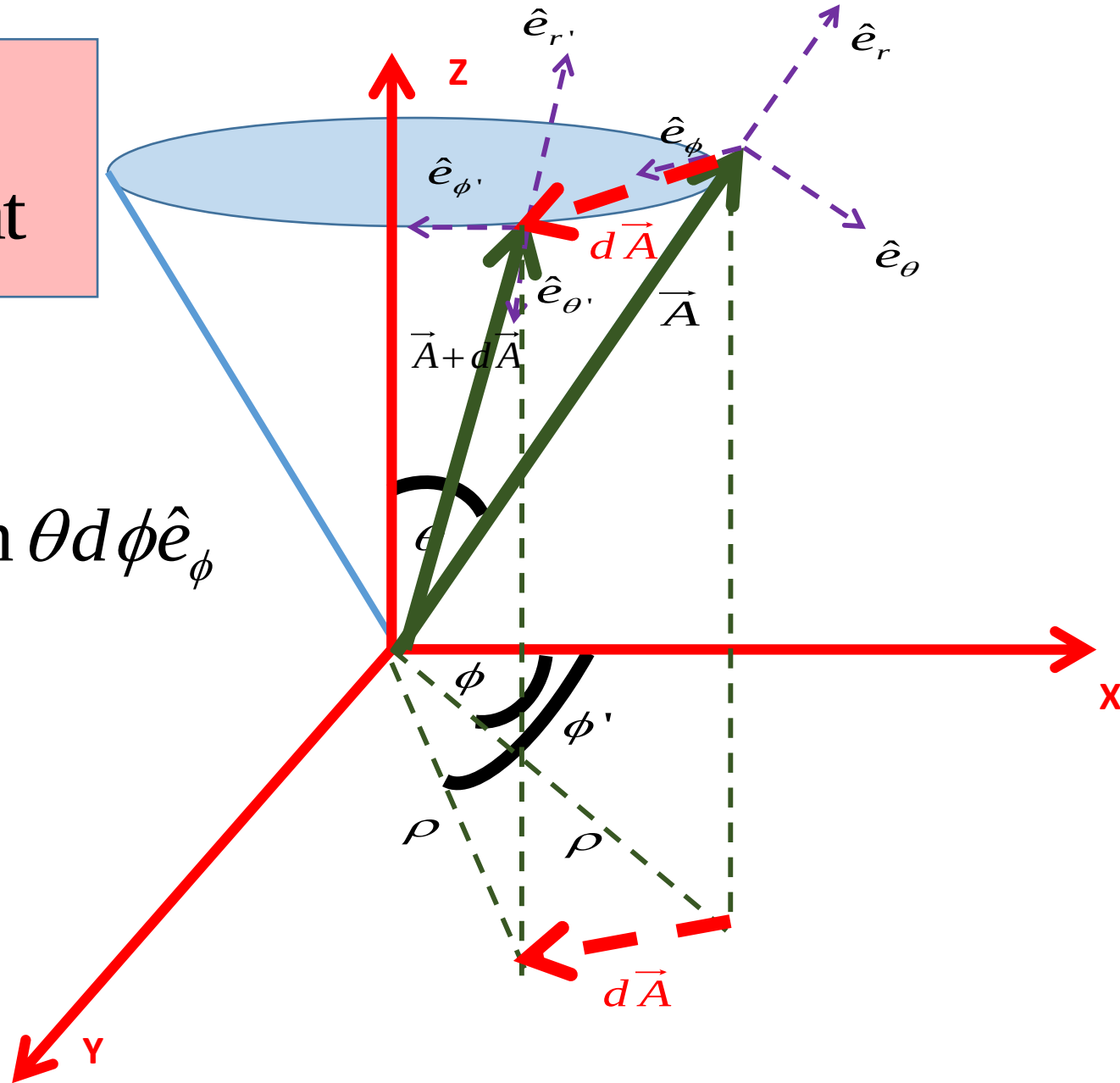


Infinitesimal line element

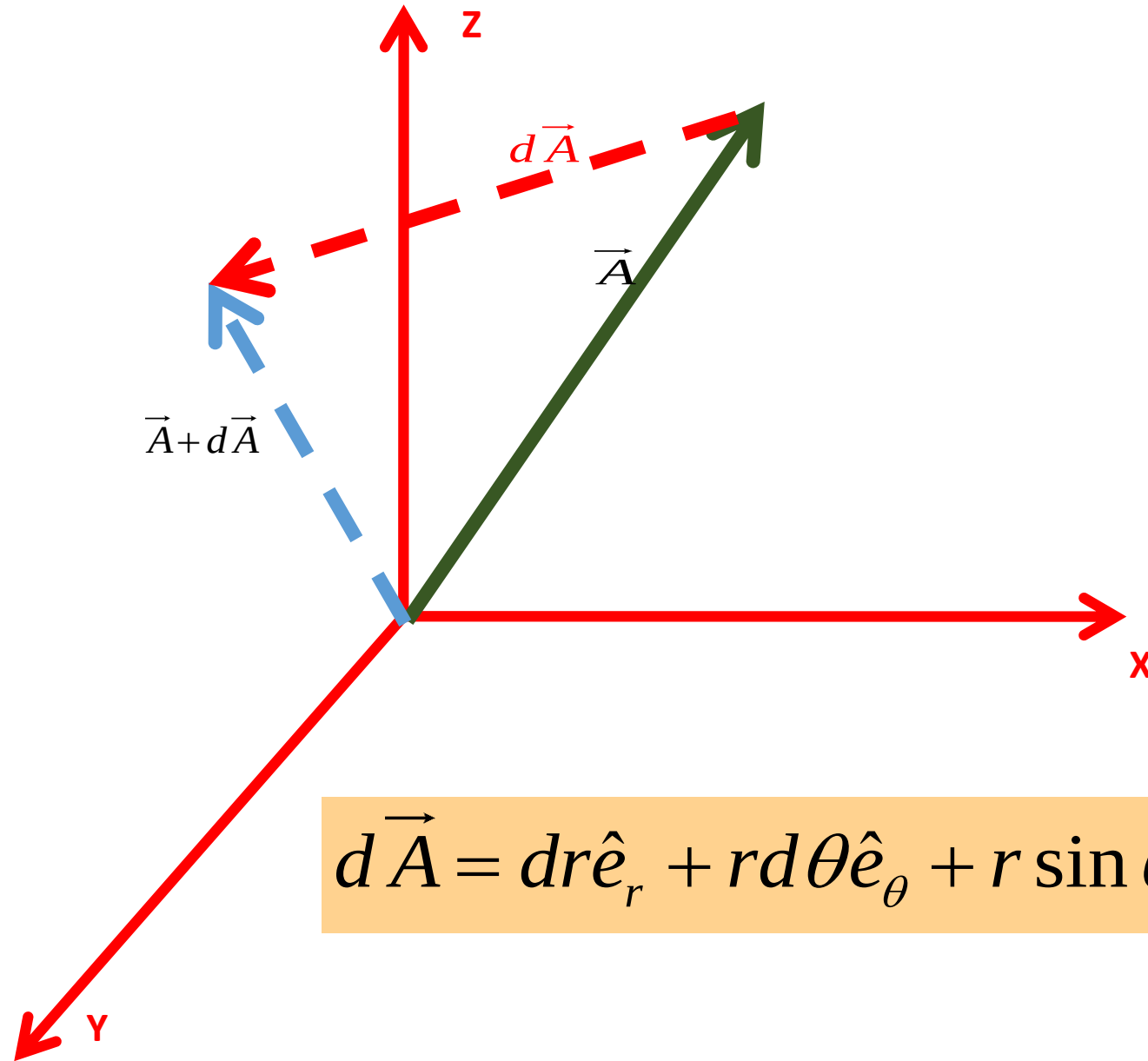
Case 3:

r and θ is constant

$$d\vec{A} = r \sin \theta d\phi \hat{e}_\phi$$



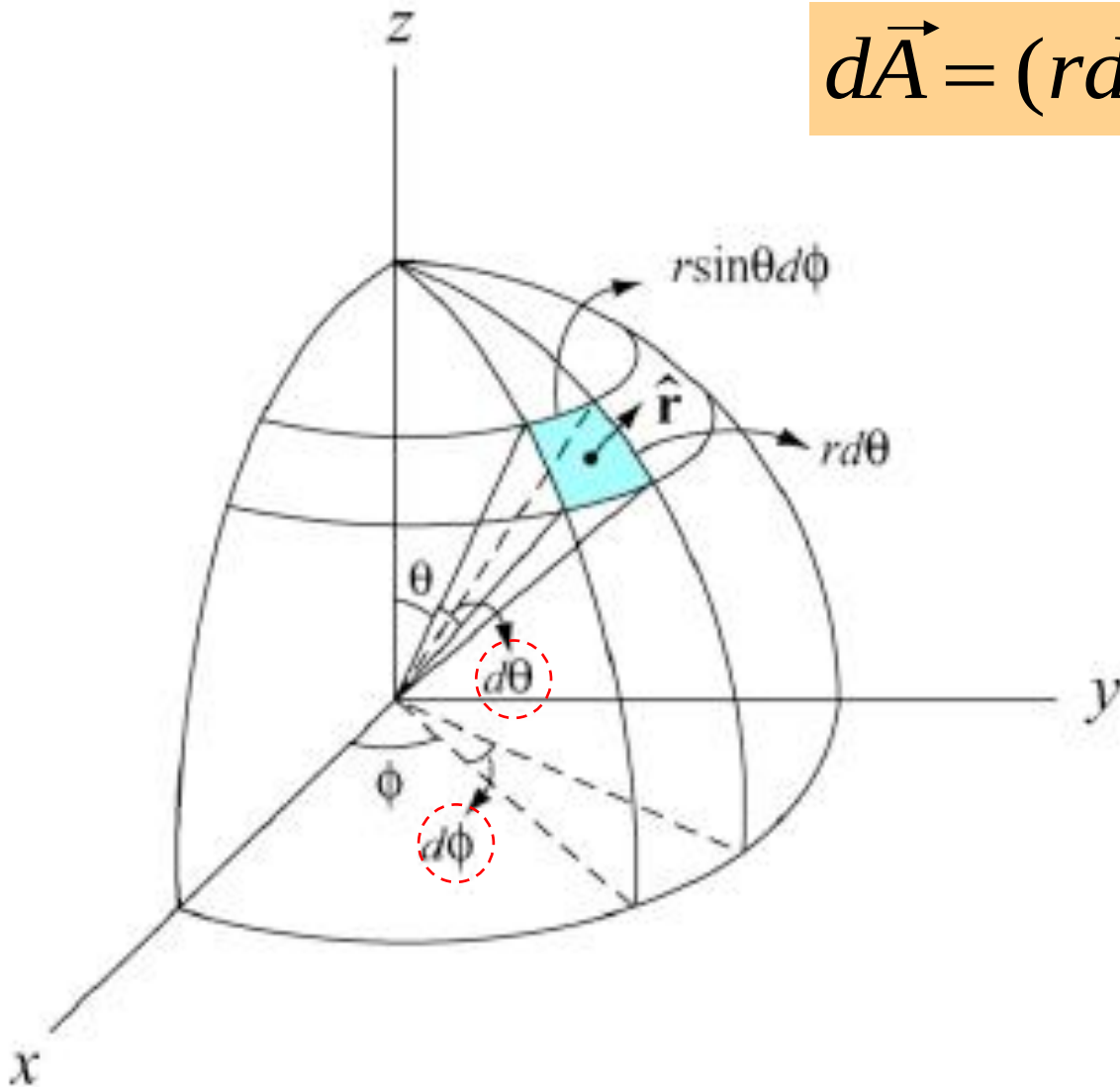
Infinitesimal line element



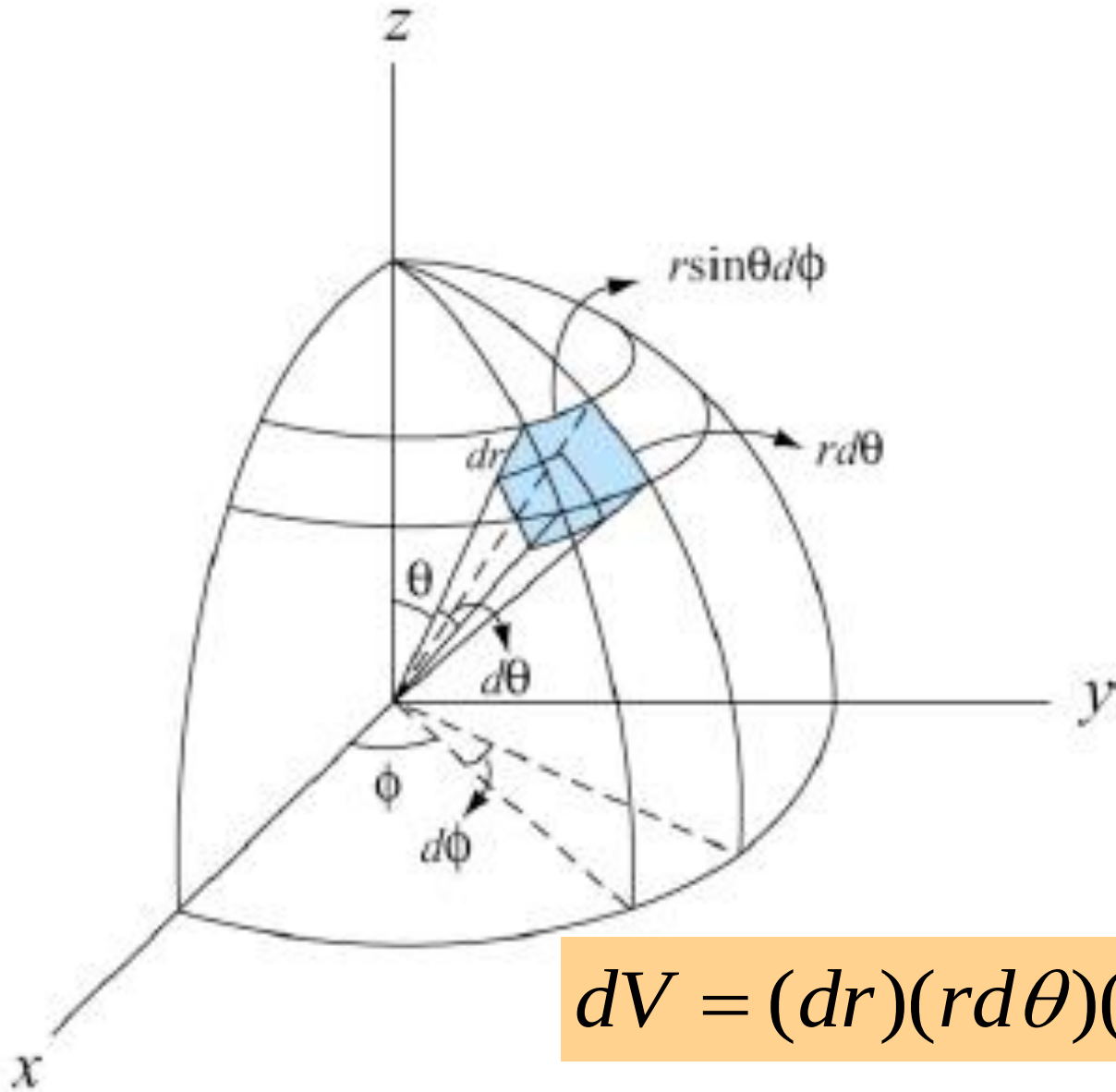
$$d\vec{A} = dr\hat{e}_r + r d\theta\hat{e}_\theta + r \sin\theta d\phi\hat{e}_\phi$$

Infinitesimal area element

$$d\vec{A} = (rd\theta)(r \sin \theta d\phi) = r^2 \sin \theta d\theta d\phi \hat{e}_r$$



Infinitesimal volume element



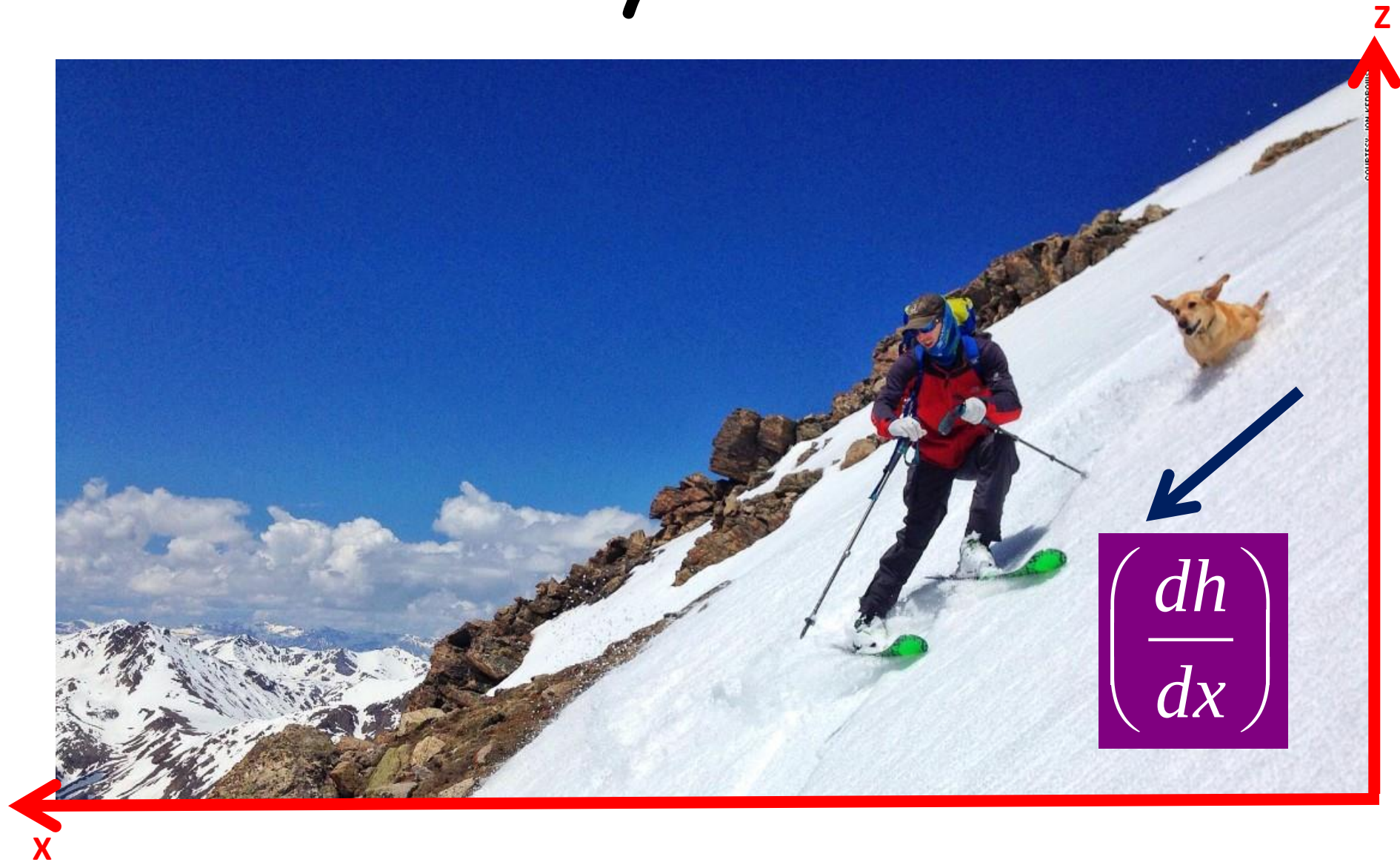
$$dV = (dr)(r d\theta)(r \sin \theta d\phi) = r^2 \sin \theta dr d\theta d\phi$$

Introduction to Vector Operators

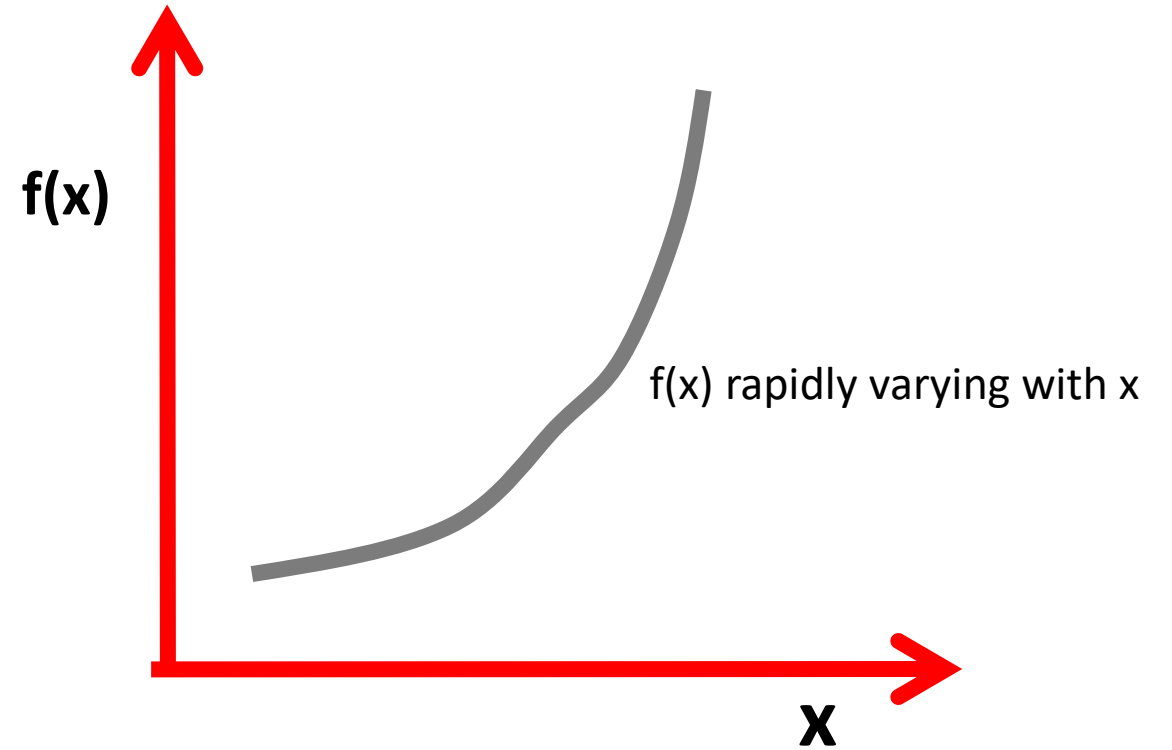
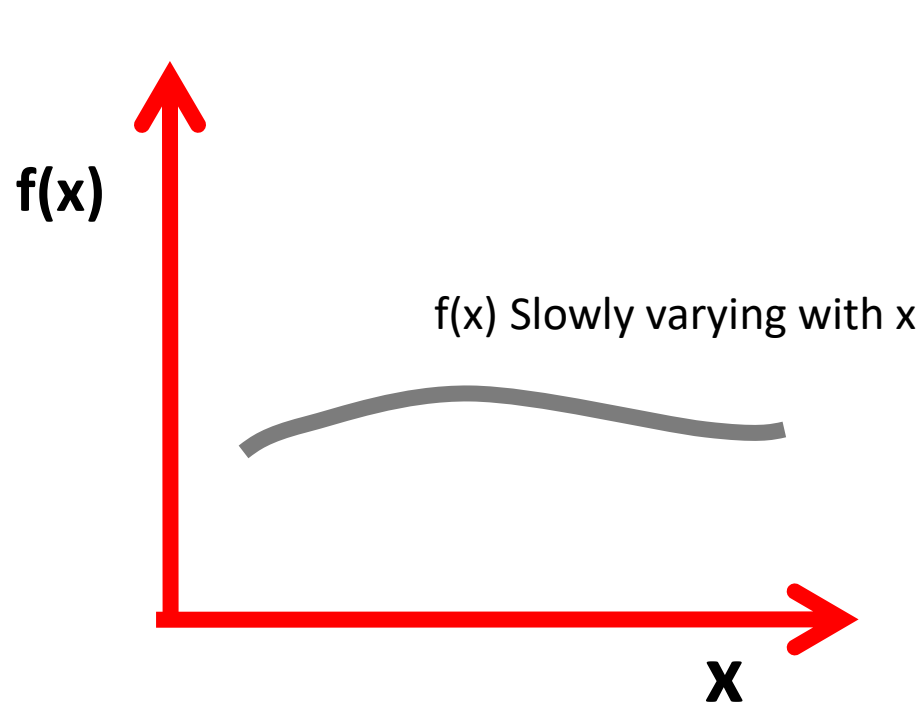
Gradient, Divergence and Curl



Ordinary Derivatives

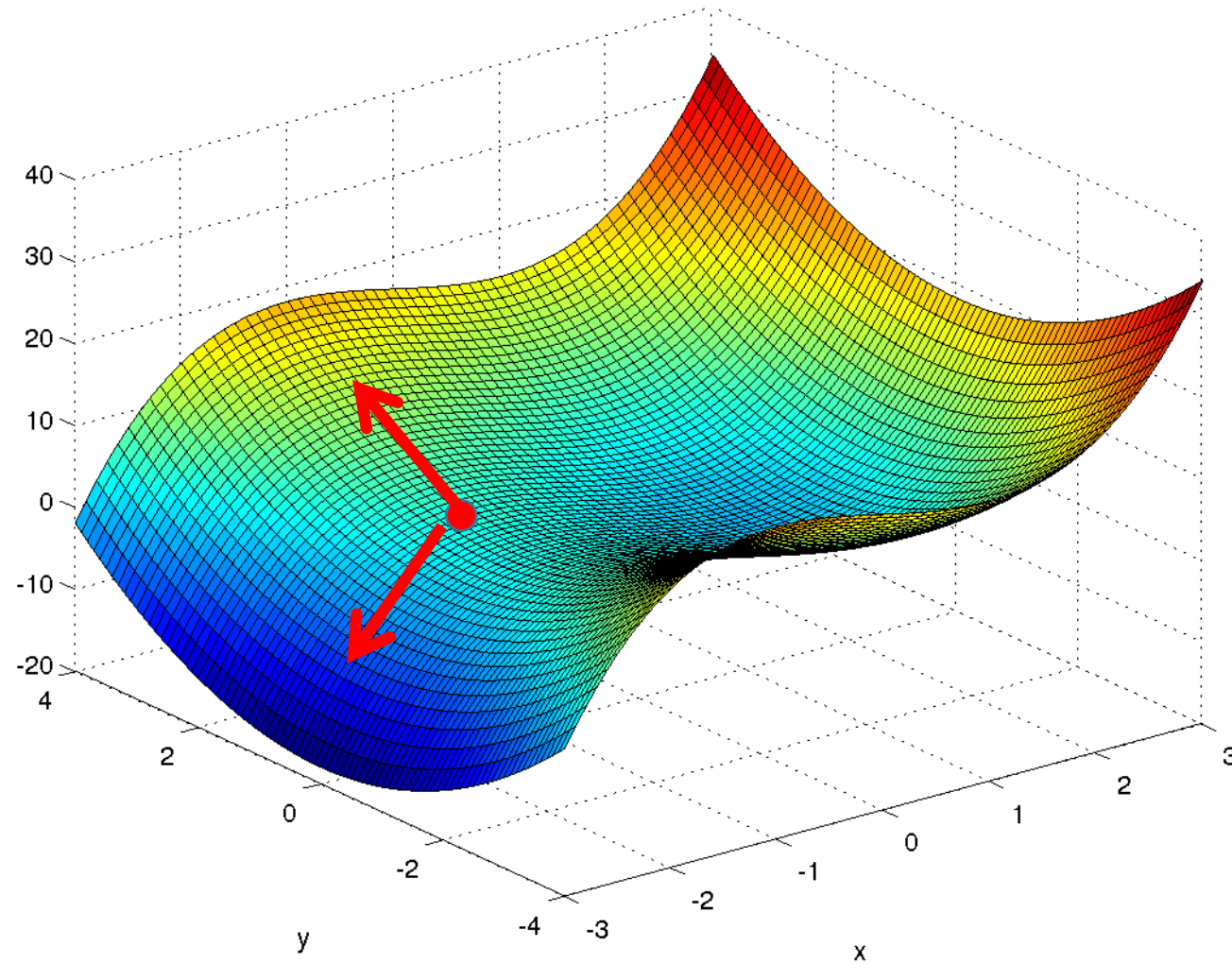


Ordinary Derivatives



$$df = \left(\frac{df(x)}{dx} \right) dx$$

Derivative of a function in 3D



Derivative is directional!

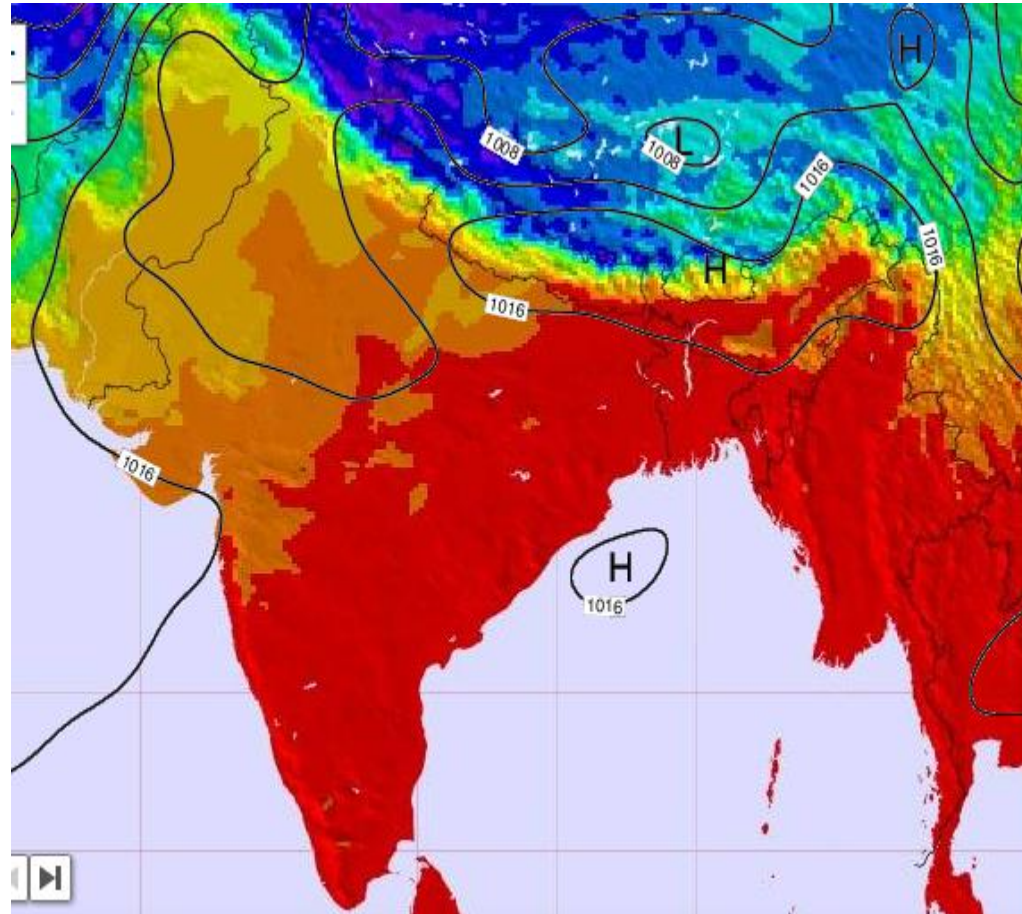
Scalar Field Function

A function of space whose value at each point is a scalar quantity.

Its value at any point is independent of where observer's frame of reference is located and how it is oriented.

Example: Temperature

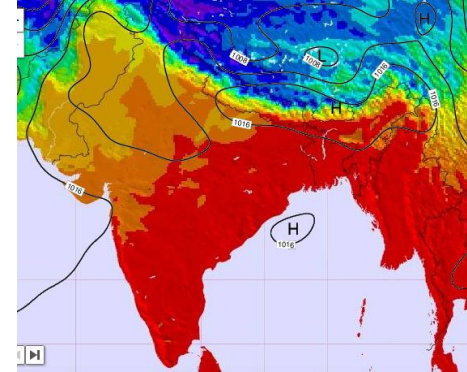
$$T(x, y, z)$$



Gradient of a scalar function

$$df = \left(\frac{df(x)}{dx} \right) dx$$

$$dT = \frac{\partial T}{\partial x} dx + \frac{\partial T}{\partial y} dy + \frac{\partial T}{\partial z} dz$$



$$T(x, y, z)$$

$$dT = \left(\frac{\partial T}{\partial x} \hat{e}_x + \frac{\partial T}{\partial y} \hat{e}_y + \frac{\partial T}{\partial z} \hat{e}_z \right) \cdot (dx \hat{e}_x + dy \hat{e}_y + dz \hat{e}_z)$$

$$dT = \vec{\nabla} T \cdot d\vec{l}$$

$$\vec{\nabla} T = \left(\frac{\partial T}{\partial x} \hat{e}_x + \frac{\partial T}{\partial y} \hat{e}_y + \frac{\partial T}{\partial z} \hat{e}_z \right)$$



Gradient

Gradient of a scalar function in different coordinate system

$$T(x, y, z)$$

$$dT = \frac{\partial T}{\partial x} dx + \frac{\partial T}{\partial y} dy + \frac{\partial T}{\partial z} dz$$

This definition is independent of the coordinate system used

$$dT = \vec{\nabla} T \bullet d\vec{l}$$

$$T(\rho, \phi, z)$$

$$dT = \frac{\partial T}{\partial \rho} d\rho + \frac{\partial T}{\partial \phi} d\phi + \frac{\partial T}{\partial z} dz$$

$$T(r, \theta, \phi)$$

$$dT = \frac{\partial T}{\partial r} dr + \frac{\partial T}{\partial \theta} d\theta + \frac{\partial T}{\partial \phi} d\phi$$

Gradient in cylindrical polar coordinate system

$$dT = \frac{\partial T}{\partial \rho} d\rho + \frac{\partial T}{\partial \phi} d\phi + \frac{\partial T}{\partial z} dz$$

$$T(\rho, \phi, z)$$

$$dT = \vec{\nabla} T \bullet d\vec{l}$$

$$d\vec{l} = d\rho \hat{e}_\rho + \rho d\phi \hat{e}_\phi + dz \hat{e}_z$$

$$dT = (\vec{\nabla} T) \cdot (d\rho \hat{e}_\rho + \rho d\phi \hat{e}_\phi + dz \hat{e}_z)$$

$$\vec{\nabla} T = \left(\frac{\partial T}{\partial \rho} \hat{e}_\rho + \frac{1}{\rho} \frac{\partial T}{\partial \phi} \hat{e}_\phi + \frac{\partial T}{\partial z} \hat{e}_z \right)$$

← Gradient

Gradient in cylindrical polar coordinate system

$$dT = \frac{\partial T}{\partial \rho} d\rho + \frac{\partial T}{\partial \phi} d\phi + \frac{\partial T}{\partial z} dz$$

$$T(\rho, \phi, z)$$

$$dT = \vec{\nabla} T \cdot d\vec{l}$$

$$\vec{\nabla} T = \left(\frac{\partial T}{\partial \rho} \hat{e}_\rho + \frac{1}{\rho} \frac{\partial T}{\partial \phi} \hat{e}_\phi + \frac{\partial T}{\partial z} \hat{e}_z \right)$$

Gradient

$$dT = \left(\frac{\partial T}{\partial \rho} \hat{e}_\rho + \frac{1}{\rho} \frac{\partial T}{\partial \phi} \hat{e}_\phi + \frac{\partial T}{\partial z} \hat{e}_z \right) \cdot (d\rho \hat{e}_\rho + \rho d\phi \hat{e}_\phi + dz \hat{e}_z)$$

$$\vec{\nabla} = \left(\hat{e}_\rho \frac{\partial}{\partial \rho} + \hat{e}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \hat{e}_z \frac{\partial}{\partial z} \right)$$

Gradient in spherical polar coordinate system

$$dT = \frac{\partial T}{\partial r} dr + \frac{\partial T}{\partial \theta} d\theta + \frac{\partial T}{\partial \phi} d\phi$$

$$T(r, \theta, \phi)$$

$$dT = \vec{\nabla} T \bullet d\vec{l}$$

$$d\vec{l} = dr\hat{e}_r + r d\theta\hat{e}_\theta + r \sin \theta d\phi\hat{e}_\phi$$

$$dT = (\vec{\nabla} T) \cdot (dr\hat{e}_r + r d\theta\hat{e}_\theta + r \sin \theta d\phi\hat{e}_\phi)$$

$$\vec{\nabla} T = \left(\frac{\partial T}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{e}_\phi \right)$$

← Gradient

Gradient in spherical polar coordinate system

$$dT = \frac{\partial T}{\partial r} dr + \frac{\partial T}{\partial \theta} d\theta + \frac{\partial T}{\partial \phi} d\phi$$

$$T(r, \theta, \phi)$$

$$dT = \vec{\nabla} T \cdot d\vec{l}$$

$$dT = \left(\frac{\partial T}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{e}_\phi \right) \cdot (dr \hat{e}_r + r d\theta \hat{e}_\theta + r \sin \theta d\phi \hat{e}_\phi)$$

$$\vec{\nabla} = \left(\hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{e}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \right)$$

$\vec{\nabla}$ in different coordinate system

**Cartesian
Coordinate system**

$$\vec{\nabla} = \left(\hat{e}_x \frac{\partial}{\partial x} + \hat{e}_y \frac{\partial}{\partial y} + \hat{e}_z \frac{\partial}{\partial z} \right)$$

**Cylindrical polar
Coordinate
system**

$$\vec{\nabla} = \left(\hat{e}_\rho \frac{\partial}{\partial \rho} + \hat{e}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \hat{e}_z \frac{\partial}{\partial z} \right)$$

**Spherical polar
Coordinate
system**

$$\vec{\nabla} = \left(\hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{e}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \right)$$

Gradient: Geometrical interpretation

$$dT = \vec{\nabla} T \bullet d\vec{l} \quad \rightarrow \quad dT = |\vec{\nabla} T| |d\vec{l}| \cos \theta$$

The gradient $\vec{\nabla} T$ points in the direction of maximum increase of the T.

If $\vec{\nabla} T = 0$ at (x,y,z) , then (x,y,z) is a stationary point of the function $T(x,y,z)$. Maxima, minima, saddle point or shoulder.

Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$

$$\begin{aligned}\nabla r &= \frac{\partial r}{\partial x} \hat{\mathbf{x}} + \frac{\partial r}{\partial y} \hat{\mathbf{y}} + \frac{\partial r}{\partial z} \hat{\mathbf{z}} \\ &= \frac{1}{2} \frac{2x}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{x}} + \frac{1}{2} \frac{2y}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{y}} + \frac{1}{2} \frac{2z}{\sqrt{x^2 + y^2 + z^2}} \hat{\mathbf{z}} \\ &= \frac{x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}} = \frac{\mathbf{r}}{r} = \hat{\mathbf{r}}.\end{aligned}$$

$$\vec{\nabla} r = \frac{\vec{r}}{r}$$

Distance from the origin increases most rapidly in the radial direction, and that its *rate* of increase in that direction is 1.